

Sequences and Factorial Notation

Precalculus Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Evaluate sequences by substituting values of n to find specific terms
- Understand and apply factorial notation including the property that $0! = 1$
- Simplify expressions involving factorials by expanding and canceling common factors

Problems

1. Write the first four terms of the sequence defined by the formula below.

$$a_n = 2n + 1$$

2. Evaluate the factorial expression below.

$$4!$$

3. Write the first four terms of the sequence defined by the formula below.

$$a_n = n^2 - 1$$

4. Evaluate the factorial expression below.

$$\frac{6!}{4!}$$

5. Write the first four terms of the sequence defined by the formula below.

$$a_n = 5 + (-1)^n$$

6. Find the 8th term of the sequence defined by the formula below.

$$a_n = 4n - 3$$



7. Simplify the factorial expression below completely.

$$\frac{8!}{2! \cdot 6!}$$

8. Write the first four terms of the sequence defined by the formula below.

$$a_n = \frac{(-1)^{n+1}}{n}$$

9. Simplify the factorial expression below completely.

$$\frac{2! \cdot 6!}{3! \cdot 5!}$$

10. Write the first four terms of the sequence defined by the formula below, then find the sum of all four terms.

$$a_n = \frac{n!}{(n+1)!}$$



Sequences and Factorial Notation — Answer Key

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Answer Key

1. Answer: 3, 5, 7, 9

- Substitute $n = 1$: $2(1) + 1 = 3$
- Substitute $n = 2$: $2(2) + 1 = 5$
- Substitute $n = 3$: $2(3) + 1 = 7$
- Substitute $n = 4$: $2(4) + 1 = 9$
- First four terms: 3, 5, 7, 9

2. Answer: 24

- Expand: $4! = 1 \times 2 \times 3 \times 4$
- Multiply step by step: $1 \times 2 = 2$, $2 \times 3 = 6$, $6 \times 4 = 24$
- $4! = 24$

3. Answer: 0, 3, 8, 15

- Substitute $n = 1$: $1^2 - 1 = 0$
- Substitute $n = 2$: $2^2 - 1 = 3$
- Substitute $n = 3$: $3^2 - 1 = 8$
- Substitute $n = 4$: $4^2 - 1 = 15$
- First four terms: 0, 3, 8, 15

4. Answer: 30

- Expand $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$ and $4! = 4 \times 3 \times 2 \times 1$
- Cancel the common $4!$ terms: $\frac{6 \times 5 \times 4!}{4!}$
- Left with $6 \times 5 = 30$

5. Answer: 4, 6, 4, 6

- Substitute $n = 1$: $5 + (-1)^1 = 5 - 1 = 4$
- Substitute $n = 2$: $5 + (-1)^2 = 5 + 1 = 6$
- Substitute $n = 3$: $5 + (-1)^3 = 5 - 1 = 4$
- Substitute $n = 4$: $5 + (-1)^4 = 5 + 1 = 6$
- First four terms: 4, 6, 4, 6

6. Answer: 29

- Substitute $n = 8$ into the formula
- $a_8 = 4(8) - 3 = 32 - 3 = 29$

7. Answer: 28

- Expand $8! = 8 \times 7 \times 6!$, $2! = 2 \times 1$, and cancel $6!$
- Expression becomes $\frac{8 \times 7 \times 6!}{2! \times 6!}$
- After canceling $6!$: $\frac{8 \times 7 \times 2 \times 1}{2 \times 1} = \frac{56 \times 2}{2} = 28$



8. Answer: 1, -1/2, 1/3, -1/4

- Substitute $n = 1$: $(-1)^2 / 1 = 1$
- Substitute $n = 2$: $(-1)^3 / 2 = -1/2$
- Substitute $n = 3$: $(-1)^4 / 3 = 1/3$
- Substitute $n = 4$: $(-1)^5 / 4 = -1/4$
- First four terms: 1, -1/2, 1/3, -1/4

9. Answer: 2/3

- Expand: $2! = 2$, $6! = 6 \times 5!$, $3! = 6$, and $5!$ cancels
- Expression becomes $\frac{2 \times 6 \times 5!}{6 \times 5!}$
- After canceling $5!$ and 6 : $\frac{2 \times 6}{6 \times 3} = \frac{12}{18} = \frac{2}{3}$

10. Answer: Terms: 1/2, 1/3, 1/4, 1/5 — Sum: 77/60

- Note that $(n+1)! = (n+1) \times n!$, so $\frac{n!}{(n+1)!} = \frac{1}{n+1}$
- $n = 1$: $1/2$
- $n = 2$: $1/3$
- $n = 3$: $1/4$
- $n = 4$: $1/5$
- Sum = $1/2 + 1/3 + 1/4 + 1/5 = 30/60 + 20/60 + 15/60 + 12/60 = 77/60$

